Lab Experiment 8: Building a Simple React Router Shopping Cart Application with AI Assistance

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Learning Outcome

At the completion of this lab experiment, students should be able to:

- Create a small React application using multiple components.
- Implement navigation between pages using React Router DOM.
- Manage application state in React.
- Pass data between components using props.
- Build a simple shopping cart system.
- Prevent duplicate items from being added to the cart.
- Remove items from the cart.

- Calculate and display the total price of items in the cart.
- Apply styling and responsive layouts to a React application.
- Use AI tools to assist with coding, debugging, and UI improvements.

Introduction

Modern web applications often contain multiple pages and interactive features such as navigation systems and shopping carts. In this lab, you will build a simple React application that combines routing and state management.

The application will include multiple pages containing interesting “curious notes” related to technology, shopping, or web development. Users will also be able to browse products, add items to a shopping cart, and navigate between pages without reloading the browser.

You will use:

- React
- React Router DOM
- React state management
- Component-based architecture
- AI coding assistants for support and troubleshooting

AI Assistance Activity

During this lab, use an AI assistant such as:

- OpenAI ChatGPT
- GitHub Copilot

Use AI tools to assist with:

- Generating starter React components
- Creating routing structures

- Styling suggestions using CSS
- Debugging React errors
- Improving UI layout
- Explaining React state and props
- Generating product data arrays

Example AI Prompts

- "Generate a React Router navigation bar with Home, Products, and Cart pages."
- "Show me how to pass cart data between React components using props."
- "Create a React shopping cart that prevents duplicate items."
- "Generate CSS styling for a modern shopping cart interface."
- "Help me calculate the total price of cart items in React."

Objective

Build a small React application where users can:

- View a list of products
- Add products to a cart
- Navigate between multiple pages
- View selected items in the cart
- Remove items from the cart
- See the total price of items

Assignment Requirements

Your application must include the following:

1. **Routing Requirements**

Use React Router DOM to implement routing between at least **three webpages**.

Required Pages

1. Home Page
2. Products Page
3. Cart Page
4. 404 Not Found Page

Each Webpage Must Include

- A title reflecting the page's theme
- An image relevant to the page content
- An informative paragraph containing something curious or interesting related to the page's topic
- Proper styling using CSS

2. Navigation Requirements

Create a navigation bar that includes links to:

- Home
- Products
- Cart

The navigation bar must:

- Display on every page
- Show the current number of items in the cart
- Allow smooth navigation between routes

3. Shopping Cart Requirements

Your application must:

- Display at least **9 products**
- Include product name, image, description, and price
- Include an **"Add to Cart"** button for each product
- Prevent duplicate items from being added
- Allow users to remove items from the cart
- Display the total price of all items
- Store cart state in the top-level App component
- Pass cart data using props

Lab Description

In this exercise, you will build a simple shopping cart application using React state and routing.

The application will contain:

- **Product List Page**

  Users can:

  - View products
  - Add items to the shopping cart
  - Read curious notes about shopping technology or e-commerce

- **Cart Page**

  Users can:

  - View selected items
  - Remove products from the cart

- View the total cost of all items

- **Home Page**

  Users can:

  - Read interesting facts about React or web development
  - Navigate to the Products and Cart pages

Suggested Project Structure

```
src/
│
├── components/
│   ├── Navbar.js
│   ├── ProductCard.js
│   └── CartItem.js
│
├── pages/
│   ├── Home.js
│   ├── Products.js
│   ├── Cart.js
│   └── NotFound.js
│
├── data/
│   └── products.js
│
├── App.js
├── App.css
└── index.js
```

Lab Procedure

Part 1 – Create the React Application

Create a new React application:

```
npx create-react-app lab
```

Install routing support:

```
npm install react-router-dom
```

Run the application for testing:

```
npm start
```

Part 2 – Configure Routing

Use:

- BrowserRouter
- Routes
- Route
- Link

Create routes for:

- /
- /products
- /cart
- * (404 page)

Part 3 – Build the Product List

Create an array containing at least 9 products.

Each product should include:

```
{
  id: 1,
  name: "Wireless Mouse",
  price: 25,
  image: "image-url"
}
```

Display products using `.map()`.

Part 4 – Add Shopping Cart Functionality

Create state in App.js:

```
const [cart, setCart] = useState([]);
```

Create functions to:

- Add items to the cart
- Remove items from the cart
- Prevent duplicates

Part 5 – Display Cart Total

Use `.reduce()` to calculate the total price.

Example:

```
const total = cart.reduce(
  (sum, item) => sum + item.price,
  0
);
```

Part 6 – Style the Application

Add CSS styling for:

- Navigation bar
- Product cards
- Buttons
- Cart layout
- Responsive design

Curious Notes Ideas

Students may include interesting facts such as:

- React uses a Virtual DOM to improve performance.
- Online shopping carts were first introduced in the 1990s.
- Single Page Applications allow navigation without full page reloads.
- React Router enables dynamic client-side routing.
- Modern e-commerce sites rely heavily on component-based UI systems.

Deliverables

Students must submit:

1. React project source code
2. Screenshots of:
    - Home page
    - Products page
    - Cart page
    - 404 page

3. A short reflection describing:
    - Challenges encountered
    - How AI assistance was used
    - What was learned during the lab

Conclusion

This lab provides hands-on experience with:

- React component architecture
- State management
- Routing
- Dynamic rendering
- Shopping cart functionality
- AI-assisted software development

By completing this activity, students will strengthen their understanding of modern front-end development practices using JavaScript and React.

Reflection Questions

In the project repository, answer the following questions in the README.md file:

1. What is the purpose of routing in React applications?
2. Why should cart state be stored in the top-level component?
3. How did you prevent duplicate cart items?
4. What role did AI tools play in your development process?
5. What improvements would you add to this application?

Grading Rubric

- Routing between pages works correctly: 20%
- Navigation bar implemented correctly: 10%
- Product list contains at least 9 items: 10%
- Add to Cart functionality works: 15%
- Remove from Cart functionality works: 10%
- Total price calculation works: 10%
- Proper styling and layout: 10%
- 404 page implemented: 5%
- Curious notes and informative content: 5%
- AI assistance reflection completed: 5%

Submission

- Complete the lab experiment and upload the project to your GitHub repository.
- Submit and share the GitHub repository link with the lab instructor.